

Assessment of antioxidant activity by using different in vitro methods.



In this study, six common tests for measuring antioxidant activity were evaluated by comparing four antioxidants and applying them to beverages (tea and juices): Trolox equivalent antioxidant capacity assay (TEAC I-III assay), Total radical-trapping antioxidant parameter assay (TRAP assay), 2,2-diphenyl-1-picrylhydrazyl assay (DPPH assay), N,N-dimethyl-p-phenylendiamine assay (DMPD assay), Photochemiluminescence assay (PCL assay) and Ferric reducing ability of plasma assay (FRAP assay). The antioxidants included gallic acid representing the group of polyphenols, uric acid as the main antioxidant in human plasma, ascorbic acid as a vitamin widely spread in fruits and Trolox as water soluble vitamin E analogue. The six methods presented can be divided into two groups depending on the oxidising reagent. Five methods use organic radical producers (TEAC I-III, TRAP, DPPH, DMPD, PCL) and one method works with metal ions for oxidation (FRAP). Another difference between these tests is the reaction procedure. Three assays use the delay in oxidation and determine the lag phase as parameter for the antioxidant activity (TEAC I, TRAP, PCL). They determine the delay of radical generation as well as the ability to scavenge the radical. In contrast, the assays TEAC II and III, DPPH, DMPD and FRAP analyse the ability to reduce the radical cation (TEAC II and III, DPPH, DMPD) or the ferric ion (FRAP). The three tests acting by radical reduction use preformed radicals and determine the decrease in absorbance while the FRAP assay measures the formed ferrous ions by increased absorbance. Gallic acid was the strongest antioxidant in all tests with exception of the DMPD assay. In contrast, uric acid and ascorbic acid showed low activity in some assays. Most of the assays determine the antioxidant activity in the micromolar range needing minutes to hours. Only one assay (PCL) is able to analyse the antioxidant activity in the nanomolar range. Black currant juice showed highest antioxidant activity in all tests compared to tea, apple juice and tomato juice. Despite these differences, results of these in vitro assays give an idea of the protective efficacy of secondary plant products. It is strongly recommended to use at least two methods due to the differences between the test systems investigated.